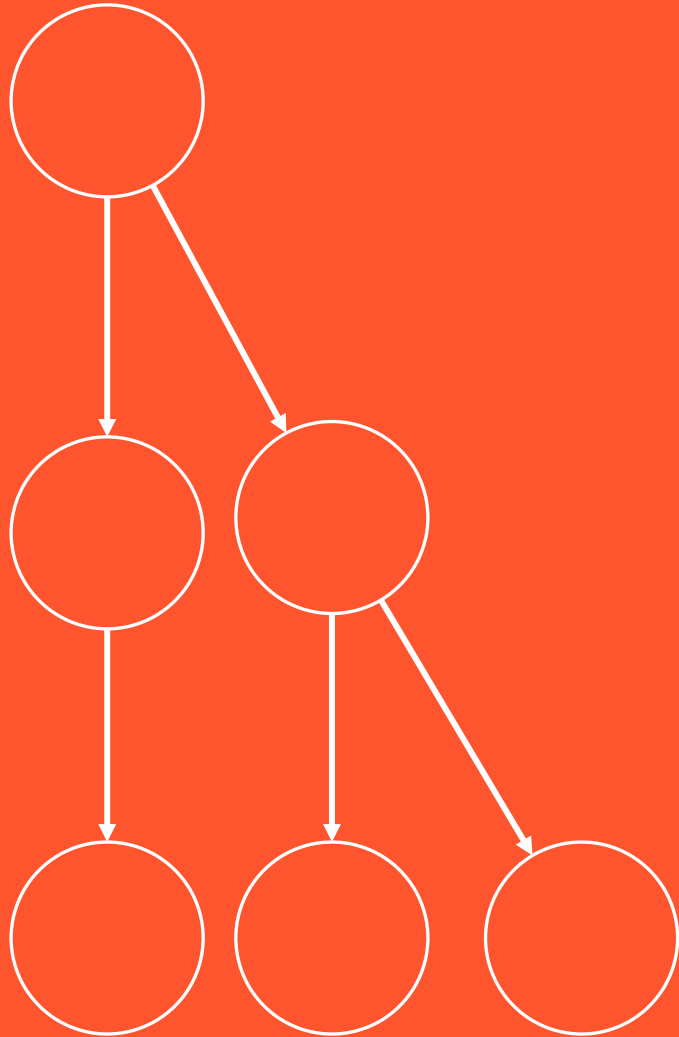




BCOG 100

Neurons and Neural Communication
Lecture 3.1 — The Evolutionary Function of Neurons



UNIVERSITY OF
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URBANA - CHAMPAIGN

One Cell, Three Questions

This is the shape of every module from here to the end of the course. One capacity, three sections in the reading, three meetings, one per level.

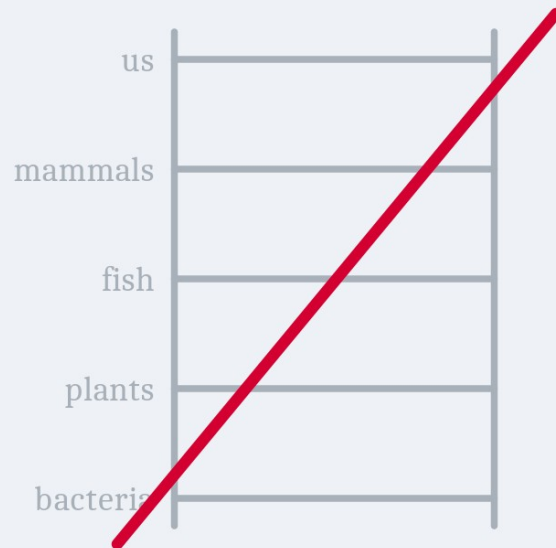
Level	The question it asks	Discipline	This module's example: the neuron
Computational	What problem is being solved, and why does solving it matter?	Evolution and ecology	Fast, long-distance, reliable signaling, for an animal that must sense, decide, and act before something else does
Algorithmic	What is represented, and by what procedure?	Cognitive psychology and AI	Receive, integrate, transmit; a threshold, and an all-or-nothing spike
Implementational	By what physical means, and at what cost?	Neuroscience	Ion gradients across a membrane, the sodium-potassium pump, and the action potential

What a Computational Question Asks About a Living Thing

Living things have no designer. So the only answer to *what is this for?* that does not appeal to magic is a **historical** one — which makes asking what a neuron is for the same as asking **what selective pressure favored building one.**

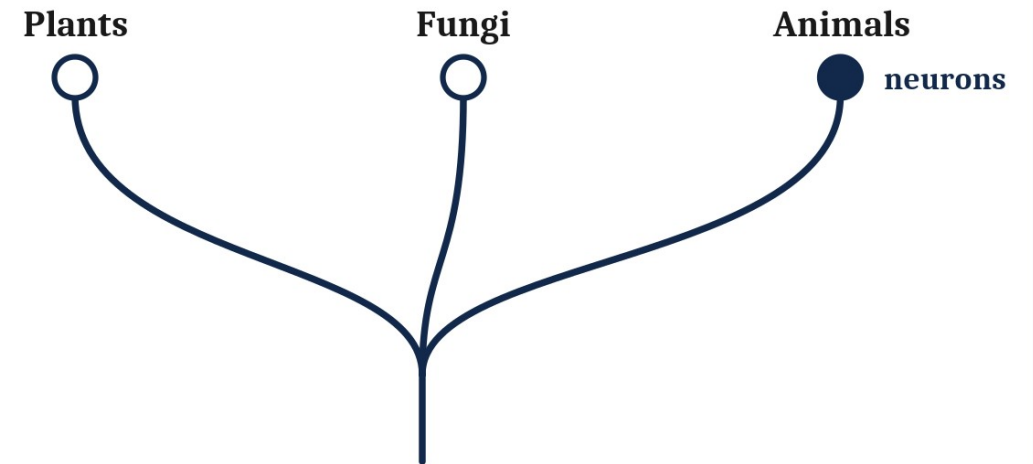
The walk forward through time is not a walk upward

Not this



A ladder, with us on the top rung

This



One eukaryotic beginning, three ways of making a living.

Only one branch built neurons. The other two are not behind.

Before There Were Cells

The RNA-world hypothesis. RNA carries information in its sequence and, folded into certain shapes, accelerates chemical reactions. One molecule, two jobs — and with it inheritance, variation and selection, before there were cells at all.

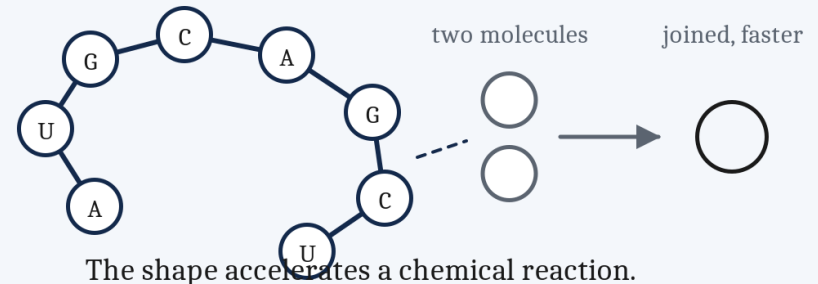
One molecule, two jobs

As a sequence



The order of the units is the message.
It carries information, as DNA does.

Folded into a shape



The shape accelerates a chemical reaction.
Catalytic RNA molecules are called ribozymes.

Inheritance

copies resemble their templates

Variation

copying is imperfect

Selection

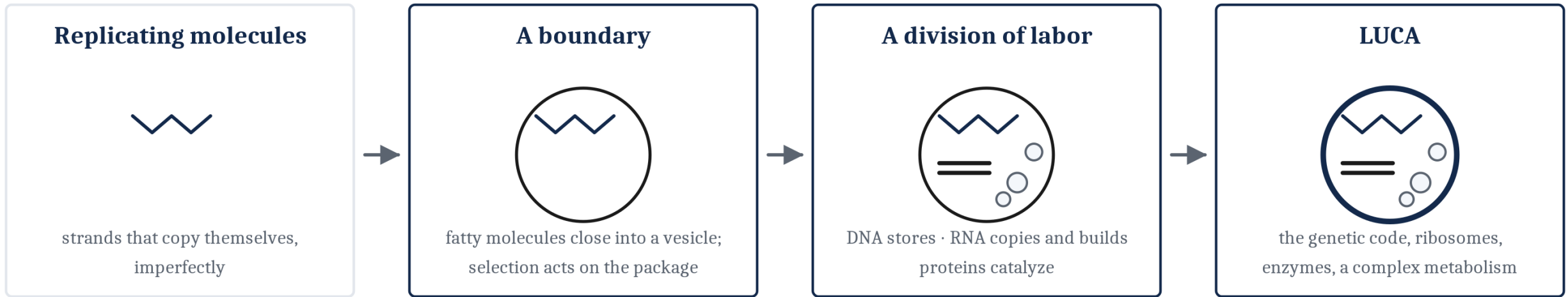
some variants do better than others

The ingredients of Darwinian evolution, present before anything anyone would call an organism.

A Boundary, and What It Bought

Replicating molecules are not yet cells. What they lack is a **boundary** — and once a vesicle could grow and split, selection could act on the whole package rather than on naked molecules.

Complexity was not assembled in one step. It accumulated.

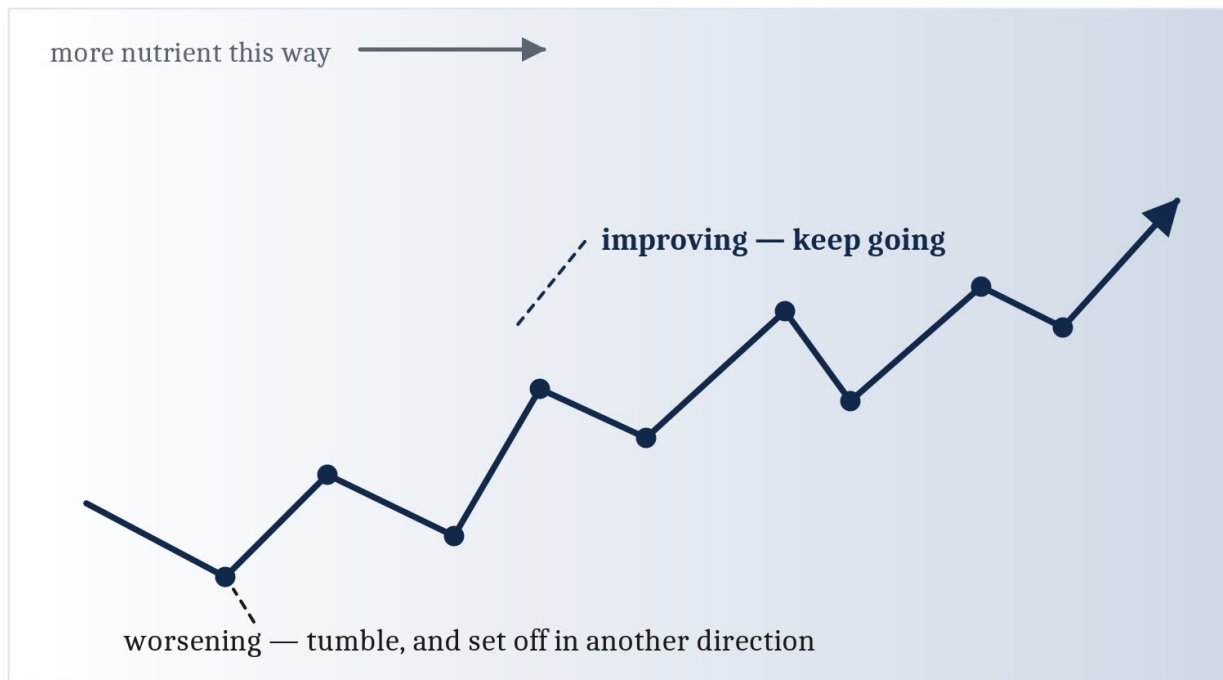


LUCA is not the first life, and not necessarily one organism.
It is the ancestral population all cellular life alive today descends from —
which means considerable evolution had already happened before it.

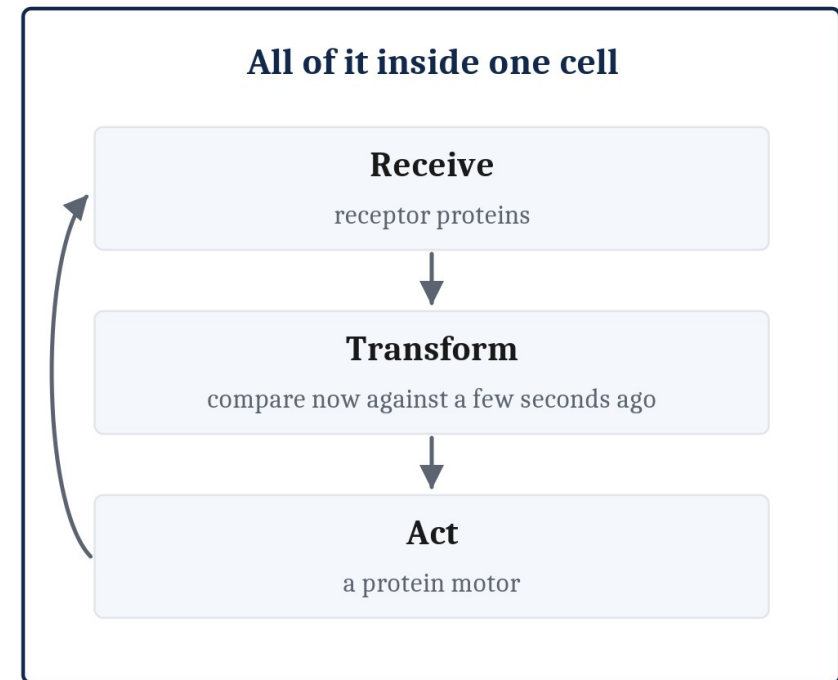
Behavior Is Older Than Neurons

A swimming bacterium compares the nutrient concentration now against the concentration a few seconds ago. **Neurons invented neither information processing nor behavior.** What they added is a way to move information fast between distant parts of one body.

The loop is billions of years older than neurons



A swimming bacterium, working up a gradient

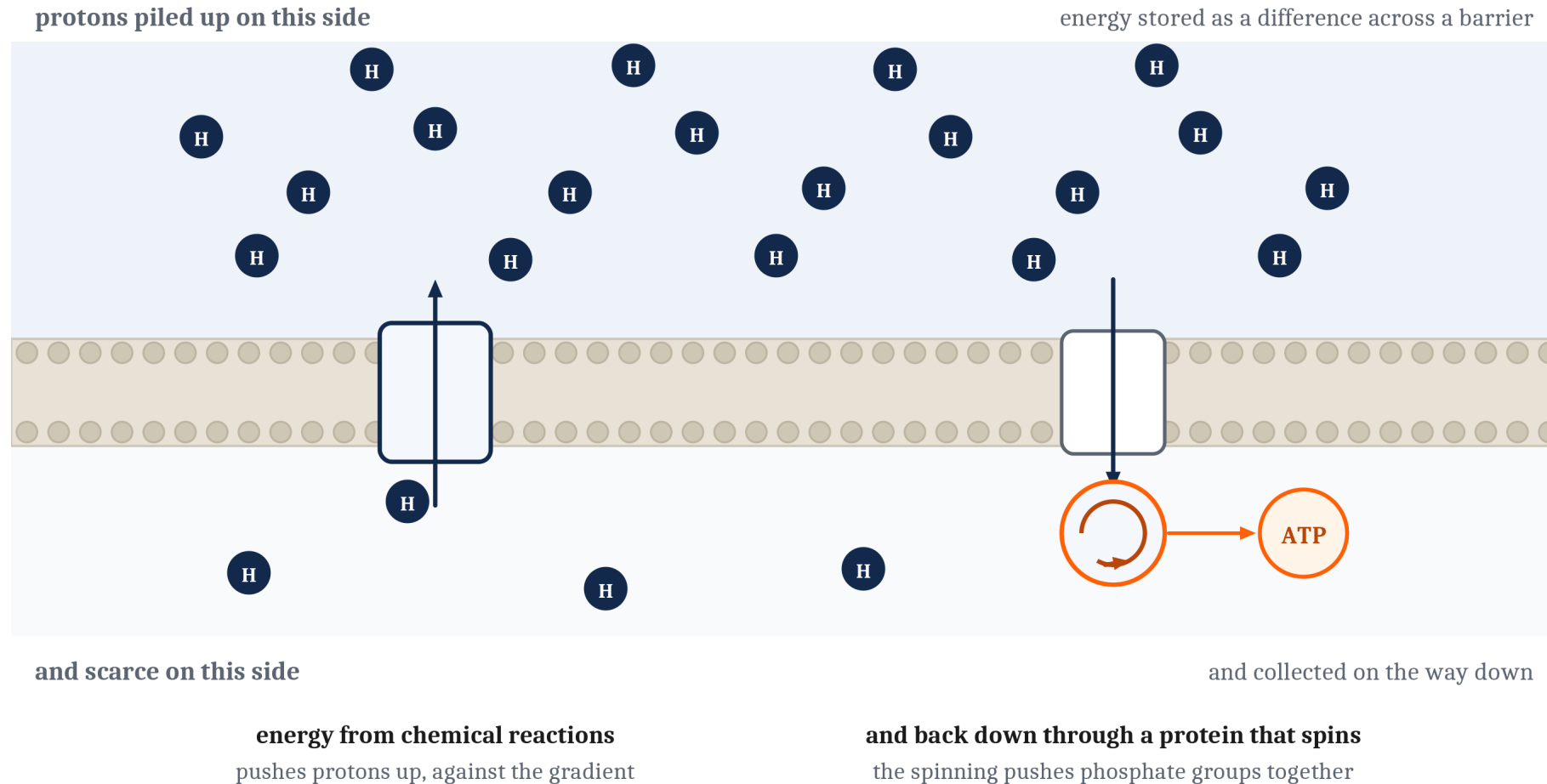


Neurons invented neither information processing nor behavior

Everything Costs, and It Is All Paid in One Currency

ATP stores energy in a chemical bond and releases it where it is needed. A cell's whole economy is making ATP and spending it.

Early life could produce energy. **It was energy poor.**



Poison Into Windfall

Photosynthesis turned sunlight into storable sugar. Its waste product, oxygen, was at first a catastrophe — the **Great Oxygenation Event**. The same aggressiveness that made oxygen a poison makes it an unusually good acceptor of electrons.

The same molecule of sugar, before and after oxygen



Oxygen was at first a catastrophe. The same chemical aggressiveness that made it a poison makes it an unusually good acceptor of electrons.

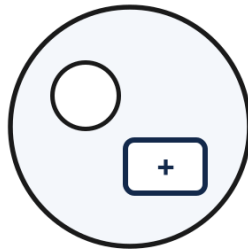
Everything expensive that life built afterward was paid for out of that windfall.

Nothing Was Waiting

A surplus makes arrangements *affordable* that were not affordable before — and *affordable* invites the misreading most people actually hold. **Nothing was waiting. There were no plans.**

The same variation, in two energy regimes

Before oxygen

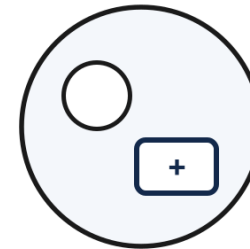


one extra internal compartment

costs more energy than it returns

the cells carrying it leave fewer descendants — it dies out

After oxygen



one extra internal compartment

the same cost, now worth paying

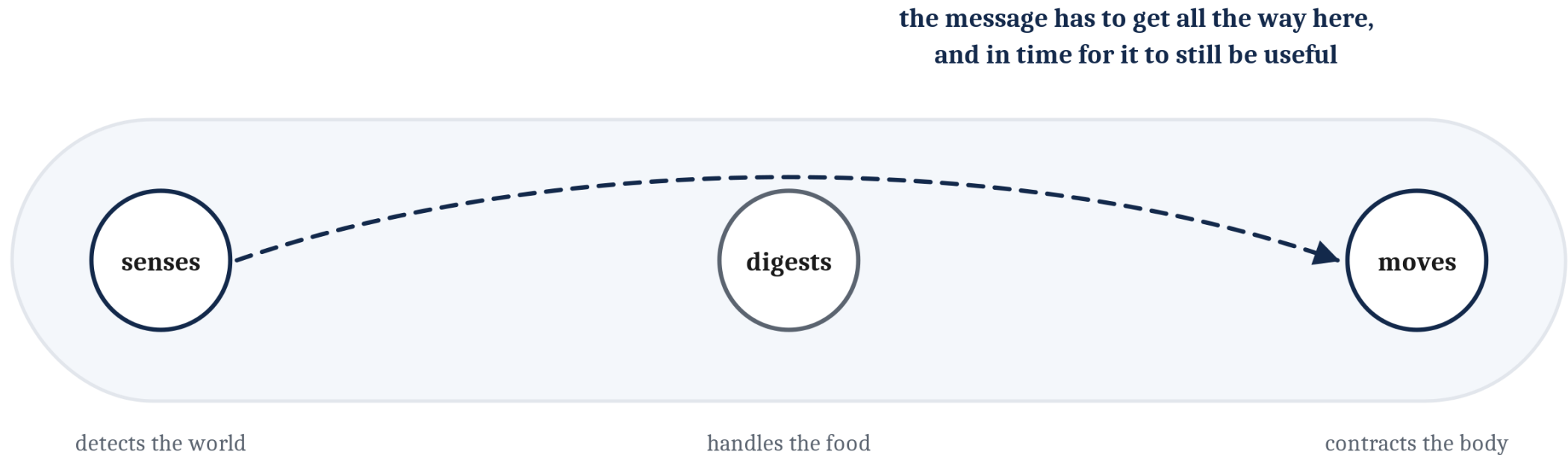
the cells carrying it leave more descendants — it spreads

Nothing about the mutation changed. What changed is the energy available to pay for it.

Selection never looked ahead. It only ever compared the variants that already existed.

Specialization Creates a Problem

Multicellularity allows **division of labor**, and division of labor creates a problem that did not exist before it. Call it the **coordination problem** — and how urgently it needs solving depends entirely on how the organism makes its living.



The cells that must act on what a sensor detects cannot detect it themselves.

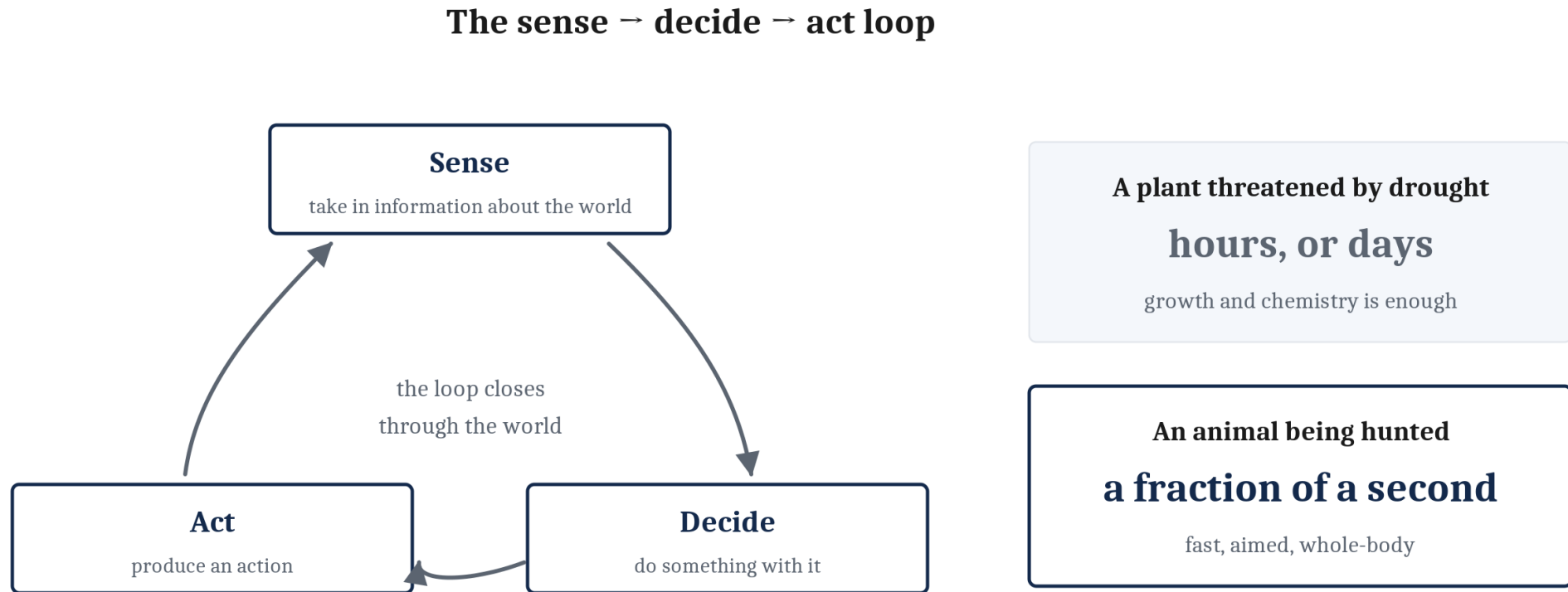
Three Ways to Be Complex

Complex multicellular life did not solve the coordination problem once. It solved it at least three times, differently — and all three are complex, all three are multicellular, all three descend from the same eukaryotic beginning. **Only one of them evolved neurons.**

	Plants	Fungi	Animals
How it gets energy	Makes its own food from sunlight	Grows into its food and absorbs it	Finds food, and eats other organisms
What the body does	Stays put; grows toward light and water	Stays put; grows through the material it feeds on	Moves through the world, and is moved through by others
What can go wrong in a second	Very little	Very little	Being eaten; losing prey; a bad step
Coordination it needs	Slow and chemical is enough	Slow and chemical is enough	Fast, aimed, whole-body

What Can Go Wrong in a Second

The separating variable is **speed**, and the reason is that animals eat other organisms and are eaten by them. The **sense → decide → act loop** returns at the algorithmic level, and again at the implementational level.

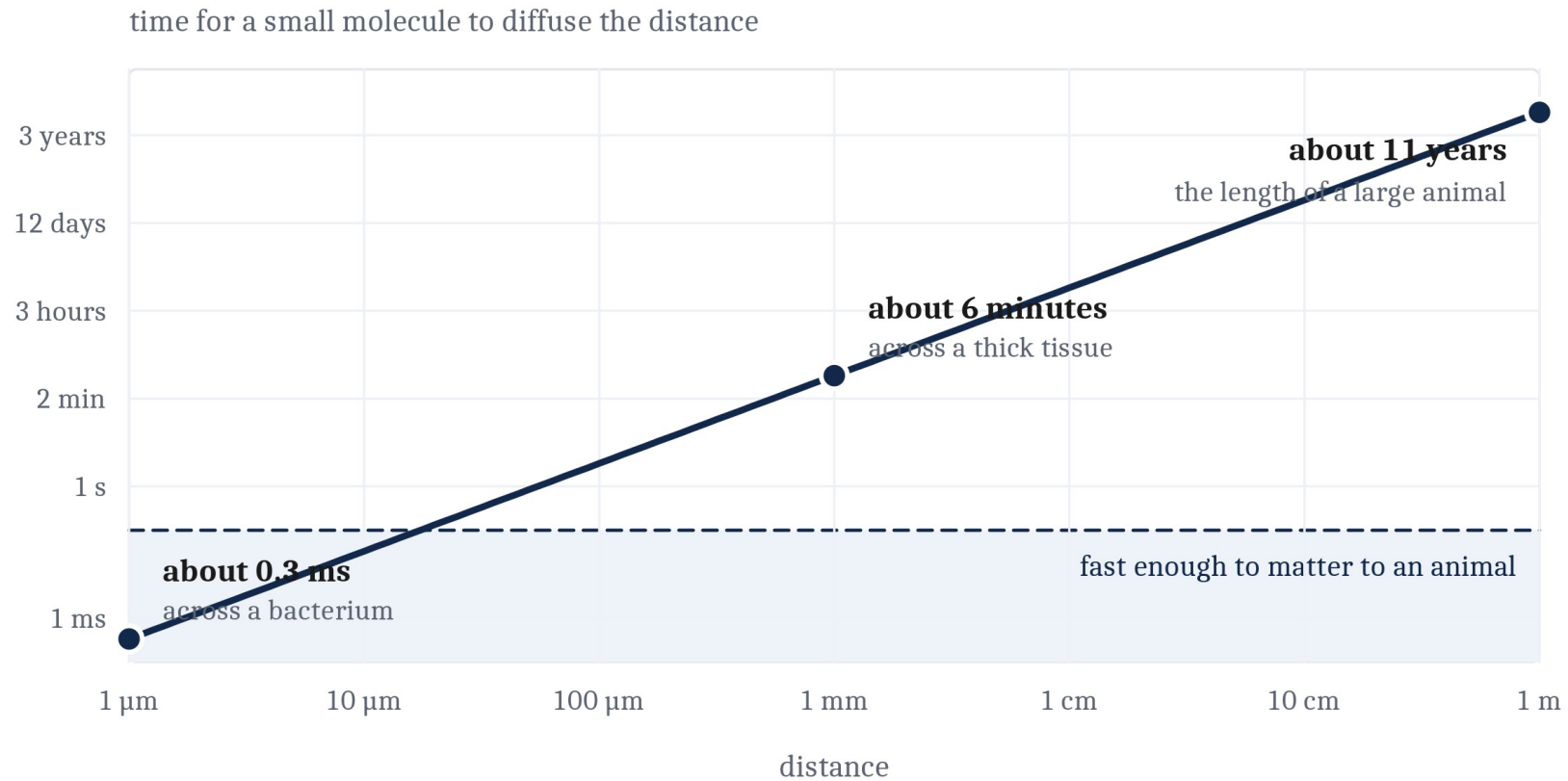


Fast enough for the action to still be the right one.

Not a problem unique to living things: any system operating on its own in a changing world faces some version of it.

Why Chemistry Was Not Fast Enough

Chemical signaling is cheap and reaches everywhere. But it moves by **diffusion**, and diffusion time grows with the **square** of the distance — so it cannot cross an animal in time, it cannot be aimed, and it cannot easily be switched off.



Both scales are logarithmic. Time grows with the square of the distance, so every doubling of distance quadruples the wait — which is why the line leaves the plate.

The Trade

A **neuron** is the cell specialized to make one trade: **give up chemical signaling's cheap broadcast reach, and buy speed, distance, and aim in return.** That is the computational-level answer to what a neuron is for.

	Chemical signaling	Neural signaling
How the message travels	by diffusion, or carried in the bloodstream	as a signal regenerated at full strength along a fiber
Crossing the length of a large animal	years, left to diffusion	a fraction of a second
Where it arrives	everywhere at once	at one specific destination
Switching it off	it lingers	it ends with the signal
What it costs	cheap, and needs no special apparatus	expensive, and the cost is the third lecture of this module
The kind of message it suits	<i>grow, get hungry, begin reproducing</i>	<i>the left side of the body should contract, now</i>

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